

HW1

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Q1: Done Q8: Done

Q1: A.3

Let $I = [0, 1]$, $\Omega = \mathbb{R}$ and for $\alpha \in \mathbb{R}$, $A_\alpha = (\alpha - 1, \alpha + 1)$, the open interval $\{x : \alpha - 1 < x < \alpha + 1\}$.

(a)

Show that $\bigcup_{\alpha \in I} A_\alpha = (-1, 2)$ and $\bigcap_{\alpha \in I} A_\alpha = (0, 1)$.

Answer

Sketch: To prove equality we argue both directions for both the Union and Intersection proofs, i.e. $\subset$ and $\supset$.

Note: we use $\subset \equiv \subseteq$ and $\supset \equiv \supseteq$ for the purposes of set relations, as noted by Dr. Ghosh in lecture.

Union

Proof.

($\subset$) For any $\alpha \in [0, 1]$, $\alpha - 1 \in [-1, 0]$ and $\alpha + 1 \in [1, 2]$, so $A_\alpha = (\alpha - 1, \alpha + 1) \subset (-1, 2)$. Since this holds for every $\alpha \in I$,

$$\bigcup_{\alpha \in I} A_\alpha \subset (-1, 2).$$

($\supset$) Let $x \in (-1, 2)$ be given, and set $\alpha = \min(\max(x, 0), 1) \in [0, 1] = I$.

- If $x < 0$: $\alpha = 0$, and since $-1 < x < 0 < 1$, $x \in A_0 = (-1, 1)$.
- If $0 \leq x \leq 1$: $\alpha = x$, and trivially $x - 1 < x < x + 1$, so $x \in A_x$.
- If $x > 1$: $\alpha = 1$, and since $0 < 1 < x < 2$, $x \in A_1 = (0, 2)$.

In every case $x \in A_\alpha$ for some $\alpha \in I$, so $x \in \bigcup_{\alpha \in I} A_\alpha$. Hence

$$(-1, 2) \subset \bigcup_{\alpha \in I} A_\alpha.$$

Combining both containments, we conclude $\bigcup_{\alpha \in I} A_\alpha = (-1, 2)$. //

Intersection

Proof.

($\subset$) Since $0, 1 \in I$, $\bigcap_{\alpha \in I} A_\alpha \subset A_0 \cap A_1 = (-1, 1) \cap (0, 2) = (0, 1)$.

($\supset$) Let $x \in (0, 1)$ be given, and let $\alpha \in I$ be arbitrary. Since $\alpha \geq 0$, $\alpha - 1 \geq -1$. And since $0 \leq \alpha \leq 1$, it follows that $\alpha - 1 \leq 0 < x$ and $x < 1 \leq \alpha + 1$, for $x \in A_\alpha$. As $\alpha \in I$ was arbitrary, $x \in \bigcap_{\alpha \in I} A_\alpha$. Hence

$$(0, 1) \subset \bigcap_{\alpha \in I} A_\alpha.$$

Combining both containments, we conclude $\bigcap_{\alpha \in I} A_\alpha = (0, 1)$. //

(b)

Suppose $J = \{x : x \in I, x \text{ is rational}\}$. Find $\bigcup_{x \in J} A_x$ and $\bigcap_{x \in J} A_x$.

Answer

Sketch: The key is that 0 and 1 are each rational, so $0, 1 \in J$; this lets us use the results from part (a) to force the reverse containments.

Union

Proof. Since $J \subset I$,

$$\bigcup_{x \in J} A_x \subset \bigcup_{\alpha \in I} A_\alpha = (-1, 2).$$

Conversely, since $0, 1 \in J$ (both are rational numbers in I),

$$\bigcup_{x \in J} A_x \supset A_0 \cup A_1 = (-1, 1) \cup (0, 2) = (-1, 2),$$

using that the two intervals overlap on $(0, 1)$. Combining both containments,

$$\bigcup_{x \in J} A_x = (-1, 2). //$$

Intersection

Since $J \subset I$,

$$\bigcap_{x \in J} A_x \supset \bigcap_{\alpha \in I} A_\alpha = (0, 1).$$

Conversely, since $0, 1 \in J$,

$$\bigcap_{x \in J} A_x \subset A_0 \cap A_1 = (-1, 1) \cap (0, 2) = (0, 1).$$

Combining both containments,

$$\bigcap_{x \in J} A_x = (0, 1). //$$

Q2: A.5

Show that $X \equiv \{0, 1\}^{\mathbb{N}}$, the set of all sequences $\{\omega_i\}_{i \in \mathbb{N}}$ where each $\omega_i \in \{0, 1\}$, is **uncountable**. Conclude that $\mathcal{P}(\mathbb{N})$ is uncountable.

Answer

Sketch: To prove $X \equiv \{0, 1\}^{\mathbb{N}}$ is uncountable, and consequently $\mathcal{P}(\mathbb{N})$ is uncountable, we must first establish (1): X is uncountable and then (2): $\mathcal{P}(\mathbb{N})$ and X are in bijection.

(1): X is uncountable

Suppose, for contradiction, that X is countable. Since X is infinite (it contains, e.g., one sequence for every finite subset of $\mathbb{N}$), countability would mean there exists a surjection $h : \mathbb{N} \rightarrow X$ enumerating every element of X : $h(1), h(2), h(3), \dots$

Write $h(n) = (h(n)_1, h(n)_2, h(n)_3, \dots)$ for the n -th sequence in the list, with $h(n)_i \in \{0, 1\}$ its i -th coordinate.

Define $\delta = (\delta_i)_{i \in \mathbb{N}}$ by flipping the diagonal entries: $\delta_i = 1 - h(i)_i$, $i \in \mathbb{N}$.

Then $\delta \in X$, since each $\delta_i \in \{0, 1\}$. But for every $n \in \mathbb{N}$, $\delta_n = 1 - h(n)_n \neq h(n)_n$, so δ and $h(n)$ disagree in coordinate n , i.e. $\delta \neq h(n)$. This holds for **every** n , so $\delta \notin \{h(1), h(2), h(3), \dots\}$, contradicting the assumption that h is surjective (enumerates all of X).

Hence no such enumeration exists, and X is uncountable.

(2): $\mathcal{P}(\mathbb{N})$ and X are in bijection

Define $\Phi : \mathcal{P}(\mathbb{N}) \rightarrow X$ by

$$\Phi(A) = \{\omega_i\}_{i \in \mathbb{N}}, \quad \omega_i = \begin{cases} 1 & \text{if } i \in A \\ 0 & \text{otherwise} \end{cases}, \quad \forall A \in \mathcal{P}(\mathbb{N}),$$

i.e. $\Phi(A)$ is the indicator sequence of A : coordinate i is 1 if $i \in A$ and 0 otherwise. We show Φ is a bijection.

- **Injective:** if $A \neq B$, then without loss of generality there is some $i \in A \setminus B$, so $\Phi(A)_i = 1 \neq 0 = \Phi(B)_i$, hence $\Phi(A) \neq \Phi(B)$.
- **Surjective:** given any $\omega = (\omega_i)_{i \in \mathbb{N}} \in X$, let $A = \{i \in \mathbb{N} : \omega_i = 1\} \in \mathcal{P}(\mathbb{N})$. Then $\Phi(A) = \omega$ by construction.

Hence Φ is a bijection, so $|\mathcal{P}(\mathbb{N})| = |X|$.

Conclusion

Bijections preserve cardinality, so $|\mathcal{P}(\mathbb{N})| = |X|$. By Step 1, X is uncountable. Therefore $\mathcal{P}(\mathbb{N})$ is uncountable.

Remarks

This recovers the general fact $|\Omega| < |\mathcal{P}(\Omega)|$ (proved for $\Omega = \mathbb{N}$ directly in the first proof) as a special case. That argument builds, for any purported bijection $g : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$, the set $A = \{i \in \mathbb{N} : i \notin g(i)\}$, which lies in no $g(i)$: a preimage x with $g(x) = A$ would force $x \in A \iff x \notin g(x) = A$, a contradiction. Translating through Φ , this A is exactly the diagonal sequence δ constructed above, since $\delta_i = 1 - \Phi(g(i))_i$ says precisely $\delta_i = 1 \iff i \notin g(i)$. The two diagonalizations are the same argument, one on sequences and one on sets.

Q3: A.9

Apply the principle of induction to establish the following:

(a)

For each $n \in \mathbb{N}$, $\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$.

Answer

Sketch: We induct on n . The base case $n = 1$ is a direct check. For the inductive step, we split off the $(n+1)^2$ term from the sum, substitute the induction hypothesis for the rest, and factor the resulting expression — reducing it to checking a polynomial identity, $2n^2 + 7n + 6 = (n+2)(2n+3)$, which puts the sum back into the closed form with $n+1$ in place of n .

Proof. Let $P(n)$ be the statement

$$\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}.$$

Base case ($n = 1$): The left side is $\sum_{j=1}^1 j^2 = 1$. The right side is $\frac{1 \cdot 2 \cdot 3}{6} = 1$. Since LHS = RHS, $P(1)$ holds.

Inductive step: Let $n \in \mathbb{N}$ be given, and suppose $P(n)$ holds, i.e.

$$\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}.$$

We want to show $P(n+1)$ holds. We have

$$\sum_{j=1}^{n+1} j^2 = \sum_{j=1}^n j^2 + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2,$$

using the induction hypothesis. Factoring out $(n+1)$,

$$\sum_{j=1}^{n+1} j^2 = (n+1) \left[\frac{n(2n+1)}{6} + (n+1) \right] = (n+1) \cdot \frac{n(2n+1) + 6(n+1)}{6} = (n+1) \cdot \frac{2n^2 + 7n + 6}{6}.$$

Since $2n^2 + 7n + 6 = (n+2)(2n+3)$, this gives

$$\sum_{j=1}^{n+1} j^2 = \frac{(n+1)(n+2)(2n+3)}{6} = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6},$$

which is exactly the statement $P(n+1)$.

Thus $P(n) \implies P(n+1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$.

(b)

For each $n \in \mathbb{N}$, $x_1, x_2, \dots, x_k \in \mathbb{R}$,

(i)

(The binomial formula). $(x_1 + x_2)^n = \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r}$.

Answer

Sketch: We induct on n . The base case $n = 1$ is immediate. For the inductive step, we multiply the induction hypothesis by $(x_1 + x_2)$, split the result into two sums, re-index one of them so both run over the same range, and combine the coefficients using Pascal's Rule $\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}$ to land exactly on the $(n+1)$ -th binomial expansion.

Proof. Let $P(n)$ be the statement

$$(x_1 + x_2)^n = \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r}.$$

Base case ($n = 1$): The left side is $(x_1 + x_2)^1 = x_1 + x_2$. The right side is

$$\sum_{r=0}^1 \binom{1}{r} x_1^r x_2^{1-r} = \binom{1}{0} x_2 + \binom{1}{1} x_1 = x_1 + x_2.$$

Since LHS = RHS, $P(1)$ holds.

Inductive step: Let $n \in \mathbb{N}$ be given, and suppose $P(n)$ holds. We show $P(n+1)$ holds. We have

$$(x_1 + x_2)^{n+1} = (x_1 + x_2)(x_1 + x_2)^n = (x_1 + x_2) \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r}$$

by the induction hypothesis. Distributing,

$$(x_1 + x_2)^{n+1} = \sum_{r=0}^n \binom{n}{r} x_1^{r+1} x_2^{n-r} + \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r+1}.$$

Re-indexing the first sum with $r \rightarrow r-1$,

$$(x_1 + x_2)^{n+1} = \sum_{r=1}^{n+1} \binom{n}{r-1} x_1^r x_2^{n+1-r} + \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n+1-r}.$$

Separating out the $r = 0$ term from the first sum's absence and the $r = n+1$ term from the second sum's absence, and combining the overlapping terms $1 \leq r \leq n$,

$$(x_1 + x_2)^{n+1} = x_2^{n+1} + \sum_{r=1}^n \left[\binom{n}{r-1} + \binom{n}{r} \right] x_1^r x_2^{n+1-r} + x_1^{n+1}.$$

By Pascal's Rule, $\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}$, so

$$(x_1 + x_2)^{n+1} = \binom{n+1}{0} x_2^{n+1} + \sum_{r=1}^n \binom{n+1}{r} x_1^r x_2^{n+1-r} + \binom{n+1}{n+1} x_1^{n+1} = \sum_{r=0}^{n+1} \binom{n+1}{r} x_1^r x_2^{n+1-r},$$

which is exactly the statement $P(n+1)$.

Thus $P(n) \implies P(n+1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$.

(ii)

(The multinomial formula).

$$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1! r_2! \dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k},$$

where the summation extends over all $(r_1, r_2, \dots, r_k)$ such that $r_i \in \mathbb{N}$, $0 \leq r_i \leq n$, $\sum_{i=1}^k r_i = n$.

Answer

Sketch: We induct on n , holding k fixed. The base case $n = 1$ follows since exactly one r_i can equal 1. For the inductive step, we multiply the induction hypothesis by $(x_1 + \dots + x_k)$, which raises one exponent r_i by 1 in each resulting term; re-indexing by $s_i = r_i + 1$ and collecting terms with the same exponents produces a common factor of $\sum_i s_i = n + 1$ in the coefficient, giving the generalized Pascal's Rule step that yields the $(n + 1)$ -th multinomial expansion.

Proof. Fix $k \in \mathbb{N}$ and $x_1, \dots, x_k \in \mathbb{R}$. Let $P(n)$ be the statement

$$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1! r_2! \dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k},$$

where the sum extends over all $(r_1, \dots, r_k)$ such that $r_i \in \mathbb{N} \cup \{0\}$, $0 \leq r_i \leq n$, and $\sum_{i=1}^k r_i = n$.

Base case ($n = 1$): The left side is $x_1 + x_2 + \dots + x_k$. On the right side, since $\sum_{i=1}^k r_i = 1$ and each $r_i \geq 0$, exactly one r_i equals 1 and all others equal 0. The term corresponding to $r_i = 1$ is

$$\frac{1!}{0! \dots 1! \dots 0!} x_1^0 \dots x_i^1 \dots x_k^0 = x_i,$$

so summing over $i = 1, \dots, k$ gives $x_1 + x_2 + \dots + x_k$. Since LHS = RHS, $P(1)$ holds.

Inductive step: Let $n \in \mathbb{N}$ be given, and suppose $P(n)$ holds. We show $P(n + 1)$ holds. We have

$$(x_1 + \dots + x_k)^{n+1} = (x_1 + \dots + x_k)^n (x_1 + \dots + x_k) = \left(\sum_{r_1 + \dots + r_k = n} \frac{n!}{r_1! \dots r_k!} x_1^{r_1} \dots x_k^{r_k} \right) (x_1 + \dots + x_k)$$

by the induction hypothesis. Distributing over the k terms $x_1, \dots, x_k$,

$$(x_1 + \dots + x_k)^{n+1} = \sum_{i=1}^k \sum_{r_1 + \dots + r_k = n} \frac{n!}{r_1! \dots r_k!} x_1^{r_1} \dots x_i^{r_i+1} \dots x_k^{r_k}.$$

For fixed i , re-index by setting $s_j = r_j$ for $j \neq i$ and $s_i = r_i + 1$, so that $\sum_{j=1}^k s_j = n + 1$ and $s_i \geq 1$. Since $r_i = s_i - 1$ gives $(s_i - 1)! = s_i! / s_i$,

$$\frac{n!}{r_1! \dots r_i! \dots r_k!} = \frac{n!}{s_1! \dots (s_i - 1)! \dots s_k!} = \frac{n! s_i}{s_1! \dots s_i! \dots s_k!}.$$

Thus

$$(x_1 + \dots + x_k)^{n+1} = \sum_{i=1}^k \sum_{s_1 + \dots + s_k = n+1} \frac{n! s_i}{s_1! \dots s_k!} x_1^{s_1} \dots x_k^{s_k},$$

where for each i the sum is understood to run only over indices with $s_i \geq 1$ (terms with $s_i = 0$ do not arise, since they would require $r_i = -1$). Swapping the order of summation and collecting terms with the same exponents $(s_1, \dots, s_k)$,

$$(x_1 + \dots + x_k)^{n+1} = \sum_{s_1 + \dots + s_k = n+1} \frac{n!}{s_1! \dots s_k!} \left(\sum_{i=1}^k s_i \right) x_1^{s_1} \dots x_k^{s_k}.$$

Since $\sum_{i=1}^k s_i = n+1$ for every term in the sum, this is

$$(x_1 + \dots + x_k)^{n+1} = \sum_{s_1 + \dots + s_k = n+1} \frac{(n+1)!}{s_1! \dots s_k!} x_1^{s_1} \dots x_k^{s_k},$$

which is exactly the statement $P(n+1)$. (This step is the natural generalization of Pascal's Rule to k indices.)

Thus $P(n) \implies P(n+1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$.

Q4: A.11

Show that the set $A = \{r : r \in \mathbb{Q}, r^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no l.u.b. in $\mathbb{Q}$.

Answer

Sketch: To show A is bounded above in $\mathbb{Q}$ but has no l.u.b. in $\mathbb{Q}$, we establish (1): A is bounded above, by exhibiting an explicit upper bound; and (2): no rational M can be the *least* upper bound, by supposing $M = \sup_{\mathbb{Q}}(A)$ exists and deriving a contradiction from a case split on $M^2 < 2$ versus $M^2 > 2$, in each case constructing a rational that violates either “upper bound” or “least.”

(1): A is bounded above

Proof. Let $r \in A$ be given. Thus $r^2 < 2$. If $r \geq 2$, then $r^2 \geq 4 > 2$, contradicting $r \in A$. Thus $r < 2$. By definition of an upper bound, 2 is an upper bound of A .

(2): A has no l.u.b. in $\mathbb{Q}$

Proof. Suppose to the contrary that $M = \sup_{\mathbb{Q}}(A)$ exists. Since $1 \in A$ (as $1^2 = 1 < 2$), $M \geq 1$, so M is a positive rational. Since $\sqrt{2}$ is irrational, no rational squares to 2, so $M^2 \neq 2$. Thus either $M^2 < 2$ or $M^2 > 2$; we derive a contradiction in each case.

Case $M^2 < 2$: We construct $r \in A$ with $r > M$, contradicting M as an upper bound. Let $\varepsilon = 2 - M^2 > 0$. Choose a rational δ with $0 < \delta < \min\left(1, \frac{\varepsilon}{2M+1}\right)$, and set $r = M + \delta$. Then

$$r^2 = M^2 + 2M\delta + \delta^2 < M^2 + \delta(2M+1) < M^2 + \varepsilon = 2.$$

So $r \in A$, yet $r = M + \delta > M$ — contradicting that M is an upper bound of A .

Case $M^2 > 2$: We construct a rational upper bound b of A with $b < M$, contradicting M as the *least* upper bound. Let $\varepsilon = M^2 - 2 > 0$. Choose a rational δ with $0 < \delta < \min\left(M, \frac{\varepsilon}{2M}\right)$, and set $b = M - \delta$. Then

$$b^2 = M^2 - 2M\delta + \delta^2 > M^2 - 2M\delta > M^2 - \varepsilon = 2.$$

Since $b > 0$ and $b^2 > 2$: for any $a \in A$, $a^2 < 2 < b^2$, and since $a, b > 0$, this gives $a < b$. So b is an upper bound of A , yet $b < M$ — contradicting that M is the *least* upper bound.

Both cases are contradictions, so no such $M \in \mathbb{Q}$ exists. Thus A has no l.u.b. in $\mathbb{Q}$.

Q5: A.12

Show that for any two sequences $\{x_n\}_{n \geq 1}, \{y_n\}_{n \geq 1} \subset \mathbb{R}$,

$$\begin{aligned} \liminf_{n \rightarrow \infty} x_n + \liminf_{n \rightarrow \infty} y_n &\leq \liminf_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) \\ &\leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n. \end{aligned}$$

Answer

Sketch: To establish the chain of four inequalities, we bound $x_n + y_n$ between $\inf_{k \geq n} x_k + \inf_{k \geq n} y_k$ and $\sup_{k \geq n} x_k + \sup_{k \geq n} y_k$ for every n , then pass this bound through the definitions of $\liminf$ and $\limsup$ — using that $\inf_{k \geq n} x_k$ is itself monotone increasing in n (so it *converges* to $\liminf x_n$, rather than merely bounding it) and $\sup_{k \geq n} x_k$ is monotone decreasing (converging to $\limsup x_n$). The middle inequality is then just the trivial fact that $\liminf$ never exceeds $\limsup$.

Step 1: The pointwise bound

For every $n \in \mathbb{N}$ and every $k \geq n$,

$$\inf_{k \geq n} x_k \leq x_k \leq \sup_{k \geq n} x_k, \quad \inf_{k \geq n} y_k \leq y_k \leq \sup_{k \geq n} y_k.$$

Taking $k = n$ and adding the two left- and right-hand inequalities gives, for every n , the bound

$$\inf_{k \geq n} x_k + \inf_{k \geq n} y_k \leq x_n + y_n \leq \sup_{k \geq n} x_k + \sup_{k \geq n} y_k,$$

which we refer to below as (*).

Step 2: The outer terms converge to $\liminf$ and $\limsup$

By definition, $\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} x_k$ and $\overline{\lim}_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} x_k$, and these limits exist (possibly $\pm\infty$) because $\inf_{k \geq n} x_k$ is non-decreasing in n and $\sup_{k \geq n} x_k$ is non-increasing in n . The same holds for y_n . Hence the left side of (*) converges:

$$\inf_{k \geq n} x_k + \inf_{k \geq n} y_k \longrightarrow \liminf_{n \rightarrow \infty} x_n + \liminf_{n \rightarrow \infty} y_n,$$

and the right side of (*) converges:

$$\sup_{k \geq n} x_k + \sup_{k \geq n} y_k \longrightarrow \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n.$$

Step 3: Pass (*) through $\liminf$ and $\limsup$

Since $\liminf$ and $\limsup$ are monotone with respect to termwise $\leq$, applying them to (*) gives, from the left inequality,

$$\liminf_{n \rightarrow \infty} \left(\inf_{k \geq n} x_k + \inf_{k \geq n} y_k \right) \leq \liminf_{n \rightarrow \infty} (x_n + y_n),$$

and by Step 2 the left side is an honest limit, equal to $\liminf_{n \rightarrow \infty} x_n + \liminf_{n \rightarrow \infty} y_n$. This gives

$$\lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} (x_n + y_n).$$

Symmetrically, applying lim sup to the right inequality of (*) and using Step 2 gives

$$\overline{\lim}_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n.$$

Step 4: The middle inequality

For any real sequence z_n , $\inf_{k \geq n} z_k \leq \sup_{k \geq n} z_k$ for every n ; taking the limit in n gives $\lim_{n \rightarrow \infty} z_n \leq \overline{\lim}_{n \rightarrow \infty} z_n$. Applying this with $z_n = x_n + y_n$:

$$\lim_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n).$$

Conclusion

Step 3 gives the outer two inequalities of the claim, and Step 4 gives the middle one. Chaining all three together yields

$$\lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n \leq \lim_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n.$$

Q6: A.17

Find the radius of convergence, ρ , for the powers series $A(x) \equiv \sum_{n=0}^{\infty} a_n x^n$ where

(a)

$$a_n = \frac{n}{(n+1)}, n \geq 0.$$

Answer

Sketch: Since $\rho = (\overline{\lim}_{n \rightarrow \infty} |a_n|^{1/n})^{-1}$ (Cauchy-Hadamard), it suffices to show $a_n^{1/n} \rightarrow 1$. Since $a_n \in (0, 1)$ for all $n \geq 1$ and $a_n \rightarrow 1$, we compute the limit of $\log(a_n^{1/n})$ directly using the standard fact $\log(1+t)/t \rightarrow 1$ as $t \rightarrow 0$.

For $n \geq 1$,

$$\log(a_n^{1/n}) = \frac{1}{n} \log\left(\frac{n}{n+1}\right) = -\frac{1}{n} \log\left(1 + \frac{1}{n}\right).$$

Writing $t = 1/n \rightarrow 0$ and using $\lim_{t \rightarrow 0} \frac{\log(1+t)}{t} = 1$,

$$\frac{1}{n} \log\left(1 + \frac{1}{n}\right) = \frac{1}{n^2} \cdot \frac{\log(1+1/n)}{1/n} \rightarrow 0 \cdot 1 = 0.$$

Hence $\log(a_n^{1/n}) \rightarrow 0$, so $a_n^{1/n} \rightarrow e^0 = 1$. Since this is an honest limit, $\overline{\lim}_{n \rightarrow \infty} |a_n|^{1/n} = 1$.

Thus $\rho = 1^{-1} = 1$.

(b)

$$a_n = n^p, \quad n \geq 0, \quad p \in \mathbb{R}.$$

Answer

Sketch: The key fact is $\log n/n \rightarrow 0$, which lets us show $n^{p/n} \rightarrow 1$ for *every* fixed p — so the radius of convergence is 1 regardless of the exponent p . We justify $\log n/n \rightarrow 0$ by applying L'Hôpital's rule to the continuous function $\log x/x$ as $x \rightarrow \infty$; since the resulting limit holds over all reals $x \rightarrow \infty$, it holds in particular along the integer sequence $n \rightarrow \infty$.

Proof. For the continuous function $f(x) = \frac{\log x}{x}$, L'Hôpital's rule gives

$$\lim_{x \rightarrow \infty} \frac{\log x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0,$$

and since this limit holds as $x \rightarrow \infty$ through the reals, it holds along the integer sequence: $\frac{\log n}{n} \rightarrow 0$.

For $n \geq 1$,

$$\log(a_n^{1/n}) = \log(n^{p/n}) = p \cdot \frac{\log n}{n} \rightarrow p \cdot 0 = 0$$

for every fixed $p \in \mathbb{R}$. Hence $a_n^{1/n} = n^{p/n} \rightarrow e^0 = 1$, so $\overline{\lim}_{n \rightarrow \infty} |a_n|^{1/n} = 1$.

(For $p < 0$ the term $a_0 = 0^p$ is undefined, but the radius of convergence depends only on the tail behavior as $n \rightarrow \infty$, so this does not affect ρ .)

Thus $\rho = 1^{-1} = 1$, for every $p \in \mathbb{R}$.

(c)

$$a_n = \frac{1}{n!}, \quad n \geq 0 \quad (\text{where } 0! = 1).$$

Answer

Sketch: We show $(1/n!)^{1/n} \rightarrow 0$ using the elementary bound $n! \geq (n/e)^n$, obtained by truncating the power series for e^n to a single term. This avoids relying on Stirling's formula, and avoids the false intermediate claim $(1/n)^{1/n} \rightarrow 0$ (in fact $(1/n)^{1/n} \rightarrow 1$, which would only give $\overline{\lim} \leq 1$ — far too weak to conclude $\rho = \infty$).

Proof. For $n \geq 1$, the exponential series $e^n = \sum_{k=0}^{\infty} \frac{n^k}{k!}$ has all nonnegative terms, so it is bounded below by its $k = n$ term:

$$e^n \geq \frac{n^n}{n!} \implies n! \geq \frac{n^n}{e^n} = \left(\frac{n}{e}\right)^n.$$

Taking n -th roots (all quantities positive),

$$(n!)^{1/n} \geq \frac{n}{e}.$$

Hence

$$0 < \left(\frac{1}{n!}\right)^{1/n} = \frac{1}{(n!)^{1/n}} \leq \frac{e}{n} \rightarrow 0.$$

By the squeeze theorem, $\lim_{n \rightarrow \infty} \left(\frac{1}{n!}\right)^{1/n} = 0$, so $\overline{\lim}_{n \rightarrow \infty} |a_n|^{1/n} = 0$.

Thus $\rho = (0)^{-1} = +\infty$: the series converges for every $x \in \mathbb{R}$.

Q7: A.18

(a)

Find the Taylor series at $a = 0$ for the function $f(x) = \frac{1}{1-x}$ in $I \equiv (-1, +1)$ and show that it converges to $f(x)$ on I .

Answer

Sketch: We first find a closed form for $f^{(n)}(x)$ by induction on n , then evaluate at $x = 0$ to read off the Taylor coefficients, which turn out to be exactly those of the geometric series $\sum x^n$. To show the series converges to f , we compute the geometric partial sums explicitly and show the error term vanishes for $|x| < 1$.

Claim. For all $n \in \mathbb{N} \cup \{0\}$, $f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$.

Proof. Base case ($n = 0$): $f^{(0)}(x) = f(x) = \frac{1}{1-x} = \frac{0!}{(1-x)^{0+1}}$, so the claim holds.

Inductive step: Suppose $f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$ for some $n \geq 0$. Differentiating,

$$f^{(n+1)}(x) = \frac{d}{dx} \left[n!(1-x)^{-(n+1)} \right] = n!(n+1)(1-x)^{-(n+2)}(1) = \frac{(n+1)!}{(1-x)^{(n+1)+1}},$$

which is the claim for $n+1$. By induction, the formula holds for all $n \in \mathbb{N} \cup \{0\}$.

The Taylor series at $a = 0$. Since $f^{(n)}(0) = n!$ for all n , the Taylor series is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)^n = \sum_{n=0}^{\infty} \frac{n!}{n!} x^n = \sum_{n=0}^{\infty} x^n.$$

Convergence to f on $I = (-1, 1)$. For $x \in I$, let $S_N(x) = \sum_{n=0}^N x^n$ be the N -th partial sum. By the finite geometric sum formula (since $x \neq 1$),

$$S_N(x) = \frac{1-x^{N+1}}{1-x} = \frac{1}{1-x} - \frac{x^{N+1}}{1-x}.$$

Since $|x| < 1$, we have $x^{N+1} \rightarrow 0$ as $N \rightarrow \infty$, so

$$\lim_{N \rightarrow \infty} S_N(x) = \frac{1}{1-x} - 0 = f(x).$$

Hence $\sum_{n=0}^{\infty} x^n$ converges to $f(x)$ for every $x \in I = (-1, 1)$.

(b)

Find the Taylor series of $1 + x + x^2$ in $I = (1, 3)$, centered at 2.

Answer

Sketch: Since g is a degree-2 polynomial, all derivatives of order 3 and higher vanish identically, so the Taylor series centered at any point is just the 3-term Taylor polynomial itself. We compute g, g', g'' at $a = 2$ and assemble the finite sum, then verify by expanding that it reproduces $g(x)$.

Let $g(x) = 1 + x + x^2$, and consider its Taylor expansion centered at $a = 2$.

Derivatives. We compute

$$g(x) = 1 + x + x^2, \quad g'(x) = 1 + 2x, \quad g''(x) = 2, \quad g^{(n)}(x) = 0 \text{ for } n \geq 3.$$

Evaluating at $a = 2$:

$$g(2) = 1 + 2 + 4 = 7, \quad g'(2) = 1 + 4 = 5, \quad g''(2) = 2, \quad g^{(n)}(2) = 0 \text{ } (n \geq 3).$$

Taylor series. Since all derivatives of order ≥ 3 vanish, the Taylor series at $a = 2$ is the finite sum

$$g(x) = \sum_{n=0}^{\infty} \frac{g^{(n)}(2)}{n!} (x-2)^n = g(2) + g'(2)(x-2) + \frac{g''(2)}{2!} (x-2)^2 = 7 + 5(x-2) + (x-2)^2.$$

Verification. Expanding,

$$7 + 5(x-2) + (x-2)^2 = 7 + 5x - 10 + x^2 - 4x + 4 = x^2 + x + 1,$$

which recovers $g(x) = 1 + x + x^2$ exactly. Since this is a finite sum (a polynomial identity), it holds for all $x \in \mathbb{R}$, and in particular on $I = (1, 3)$.

(c)

Let

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } |x| < 1, x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

(i)

Show that f is infinitely differentiable at 0 and compute $f^{(j)}(0)$ for all $j \geq 1$.

Answer

Sketch: Away from 0, each derivative of f has the form (rational function of x) $\times e^{-1/x^2}$, since differentiating this shape via the chain/product rule always produces another term of the same shape. To handle $x = 0$, we show e^{-1/x^2} divided by any power of x still $\rightarrow 0$ as $x \rightarrow 0$ (the exponential decay dominates any polynomial blow-up), then induct on j using the definition of the derivative at a point to show every $f^{(j)}(0) = 0$.

Claim. f is infinitely differentiable at 0, and $f^{(j)}(0) = 0$ for all $j \geq 1$.

Setup. For $x \neq 0$ (with $|x| < 1$), one can show inductively that each derivative of f has the form

$$f^{(j)}(x) = \sum_{k=0}^{j-1} a_{k,j} x^{2k-3j} e^{-1/x^2}, \quad x \neq 0,$$

for suitable constants $a_{k,j}$ (this follows by repeated differentiation via the chain and product rules; each differentiation multiplies existing terms by a rational function of $1/x$ and adds new terms of the same general shape). We take this representation as given and focus on differentiability at $x = 0$.

Key Lemma. For every integer $m \geq 0$,

$$\lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^m} = 0.$$

Proof: Substituting $s = 1/|x|$ so that $s \rightarrow \infty$ as $x \rightarrow 0$, we have $\left| \frac{e^{-1/x^2}}{x^m} \right| = s^m e^{-s^2} \rightarrow 0$ as $s \rightarrow \infty$, since the exponential e^{-s^2} decays faster than any polynomial power s^m grows.

Proof (by induction on j) that $f^{(j)}(0)$ exists and equals 0 for all $j \geq 0$.

Base case ($j = 0$): By definition $f(0) = 0$. Also $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} e^{-1/x^2} = 0$ by the Lemma (with $m = 0$), so f is continuous at 0 with $f(0) = 0$.

Inductive step: Suppose $f^{(j)}(0) = 0$ and, for $x \neq 0$, $f^{(j)}(x) = \sum_{k=0}^{j-1} a_{k,j} x^{2k-3j} e^{-1/x^2}$. Then by the definition of the derivative,

$$f^{(j+1)}(0) = \lim_{h \rightarrow 0} \frac{f^{(j)}(h) - f^{(j)}(0)}{h} = \lim_{h \rightarrow 0} \frac{f^{(j)}(h)}{h} = \lim_{h \rightarrow 0} \sum_{k=0}^{j-1} a_{k,j} \frac{e^{-1/h^2}}{h^{3j-2k+1}}.$$

Each term in this finite sum tends to 0 by the Key Lemma (with $m = 3j - 2k + 1 \geq 0$). Hence the whole limit is 0, so $f^{(j+1)}(0)$ exists and equals 0.

By induction, $f^{(j)}(0)$ exists and equals 0 for every $j \geq 0$; in particular, f is infinitely differentiable at 0 and $f^{(j)}(0) = 0$ for all $j \geq 1$.

(ii)

Show that the Taylor series at $a = 0$ converges but not to f on $(-1, 1)$.

Answer

Sketch: Since every derivative of f vanishes at 0 by part (c)(i), the Taylor series at $a = 0$ is identically the zero function, which trivially converges everywhere. But $f(x) = e^{-1/x^2} > 0$ for every $x \neq 0$, so the series (constant 0) disagrees with f at every nonzero point of $(-1, 1)$.

Claim. The Taylor series of f at $a = 0$ converges on $(-1, 1)$, but not to f .

Proof. From part (c)(i), $f^{(n)}(0) = 0$ for all $n \geq 0$ (including $f(0) = 0$). Therefore the Taylor series at $a = 0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)^n = \sum_{n=0}^{\infty} \frac{0}{n!} x^n = \sum_{n=0}^{\infty} 0 = 0,$$

i.e., the Taylor series is identically the zero function. This series trivially converges (to 0) for every $x \in \mathbb{R}$, and in particular for every $x \in (-1, 1)$.

However, for any $x \in (-1, 1)$ with $x \neq 0$, we have $f(x) = e^{-1/x^2} > 0 \neq 0$. Hence the Taylor series converges to 0, but $f(x) \neq 0$, so the Taylor series does **not** converge to $f(x)$ at any such point. Only at $x = 0$ does the series' value (0) agree with $f(0) = 0$.

Thus the Taylor series of f at $a = 0$ converges on all of $(-1, 1)$, but it converges to f only at the single point $x = 0$, not on $(-1, 1)$ as a whole.

Q8: A.24 (c)

Draw the open unit ball $B_p \equiv \{x : x \in \mathbb{R}^2, d_p(x, 0) < 1\}$ in $\mathbb{R}^2$ for $p = 1, 2$ and ∞ .

Answer

$p = 1, 2$ and ∞ are shown below in black, red, and blue respectively.

